

Yili Wen

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Research Interests

I'm interested in designing and building **wearable and robotic systems that restore and augment human movement**. I am drawn to the full loop from human intent to physical assistance, and I work across **actuation and control, embedded sensing, and digital fabrication**. My approach is hands-on and human-centered. I prototype real hardware and evaluate it with the people it is built for. Looking ahead, I want to extend my research toward **assistive and rehabilitation robotics and exoskeletons**, and toward closing the loop between neural intent and adaptive physical assistance.

Education

2025 – Present	Graduate Student, Computer Science — The University of Texas at Dallas Design & Engineering for Making (DE4M) Lab, advised by Liang He. M.S. in Computer Science (Interactive Computing) expected Spring 2027 .
2021 – 2025	B.A. in Computation and Design — Duke Kunshan University Dual-degree program with B.A. in Interdisciplinary Studies, Duke University . Undergraduate research advisor: Xin Tong.

Publications

2026 [E.1]	* Yili Wen , Yafei Ge, Xin Tong, Liang He. “PuffFab: Making Shape-Transformable Rice Paper for Playful Food Fabrication.” (Work-in-Progress). <i>ACM TEI 2026</i> .
2026 [W.2]	* Yili Wen , Ceci Verbaarschot, Liang He. “From Intention to Action: An Invasive BCI-Driven Wearable Framework for Everyday Hand Assistance in SCI Patients.” (Workshop paper & poster). <i>Everyday Wearable for Personalized Health and Well-Being Workshop, ACM CHI 2026</i> .
2026 [P.1]	Georgia Loewen, Karen Anne Cochrane, Yili Wen , Audrey Girouard. “Personas at Play: Understanding Disabled Player Experience to Expand Accessible Controller Design.” (Poster). <i>Graphics Interface (GI) 2026</i> .
2025 [C.1]	* Yili Wen , Jiaxin Wang, Hongni Ye, Liang He, Min Fan, Xin Tong. “‘I Want to Use Technology to Help Farmers in the Future’: Enhancing Children’s Understanding of Farming through Tangible User Interfaces and Embodied Interaction.” <i>ACM Interaction Design and Children (IDC) 2025</i> .
2024 [C.2]	Hongni Ye, Ruoxin You, Kaiyuan Lou, Yili Wen , Xin Yi, Xin Tong. “‘I Keep Sweet Cats in Real Life, But What I Need in the Virtual World Is a Neurotic Dragon’: Virtual Pet Design with Personality Patterns.” <i>Chinese CHI 2024</i> .
2023 [W.1]	Hongni Ye, Ruoxin You, Kaiyuan Lou, Yili Wen , Xin Yi, Xin Tong. “PetGen: Design and Generation of Virtual Pets.” (Workshop). <i>IEEE ICME Workshops (ICMEW) 2023</i> .

Manuscripts In Preparation

–	* Yili Wen , Ceci Verbaarschot, Liang He. “A Soft Grasp-Assist Exoskeleton for Spinal Cord Injury: Design and Intent-Driven Control.” In preparation; targeting <i>IEEE Robotics and Automation Letters (RA-L)</i> .
–	* Yili Wen , Liang He. “Sequential vs. Simultaneous Actuation for Guiding Blind and Low-Vision Runners.” In preparation; targeting <i>ACM IMWUT</i> (Nov. 2026 cycle).
–	Tianjian Liu, Jingwei Sun, Yili Wen , Haiqi Shen, Liuxin Zhang, Yu Zhang, Qianying Wang, Xin Tong. “Human–AI Co-Writing in Knowledge-Intensive Tasks: Behaviors, Challenges, and Collaboration Patterns.” In preparation; targeting <i>Int. J. Human–Computer Interaction (IJHCI)</i> .

Selected Research Projects

2025 – Present	Soft Grasp-Assist Exoskeleton for Spinal Cord Injury (“From Intention to Action”) A wearable soft exoskeleton that restores grasping to people with spinal cord injury, driven by the wearer’s own decoded intent. - Designed soft pneumatic finger actuators (printed silicone using Form4B) and a layer-jamming wrist brace for stable, compliant grasp assistance during everyday tasks. - Built the embedded control and pressure-sensing stack, and a neural-intent (BCI) pipeline that turns decoded intention into adaptive assistance. - Ground the design in real rehabilitation needs through a co-design study with clinicians.
2025 – Present	Wearable Actuation System for Guiding Blind & Low-Vision Runners A body-worn actuation system that gives blind and low-vision runners directional turn and hazard guidance on their own body, so they can run independently instead of relying on a sighted guide. - Built distributed spinal actuator tiles (motorized press and lateral drag) that reproduce a guide’s directional pull, paired with on-device vision and a companion iOS app. - Investigating a novel sequential (cranio-caudal) vs. simultaneous actuation scheme, grounded in turning biomechanics and a formative study with BLV runners and guides.
2025	PuffFab: Shape-Transformable Rice Paper for Playful Food Fabrication (TEI ’26) A design-to-fabrication pipeline that lets people make edible rice paper puff from flat patterns into raised 2.5D forms when fried. Modified a 3D printer and built the design tool so users customize a flat pattern, using moistening and cutting to control how each region expands or curls during frying.
2024 – 2025	Adaptive Game-Controller Hardware for Upper-Limb Motor Disabilities (GI ’26) DIY, reconfigurable game controllers that let players with upper-limb motor disabilities shape the input to their own body. - Designed and fabricated the modular, remappable controller prototypes that served as the physical platform for the user study. - Co-authored the resulting poster (Graphics Interface 2026), which studied player experience and narrative-inquiry methods around this platform.
2023 – 2024	WooGu: Tangible System for Children’s Agricultural Literacy (IDC ’25) An educational game that teaches children farm-to-table knowledge through a tangible interface, sensor-embedded 3D-printed tools, and whole-body play. - Designed and built the system, pairing sensor-embedded 3D-printed farming tools with an augmented digital display to give coordinated visual, tactile, and digital feedback. - Evaluated with 15 children aged 5 to 12, where it improved their farming knowledge, engagement, and attitudes toward agriculture.

Research Experience

2025 – Present	Research Assistant — Liang He Design & Engineering for Making (DE4M) Lab, UT Dallas, USA.
2023 – 2025	Research Assistant — Xin Tong HCI Lab, Duke Kunshan University & Information Hub, HKUST (Guangzhou), China.
2023 – 2025	Research Assistant — Karen Cochrane Embodied Computing Lab, University of Waterloo, Canada.
2022 – 2023	Research Assistant — Fan Liang Center for the Study of Contemporary China, Duke Kunshan University, China.

Teaching & Outreach

Summer 2026	Instructor — TAST STEM Bridge Summer Camp Department of Computer Science, UT Dallas. Six-week camp (June–July 2026); designed and delivered hands-on lectures and build sessions.
Summer 2026	Instructor — CS Outreach Summer Camp Department of Computer Science, UT Dallas. June–July 2026.

Spring 2026	Teaching Assistant — CS 4376 Object-Oriented Design Department of Computer Science, UT Dallas.
Fall 2025	Teaching Assistant — CS 6326 Introduction to Human-Computer Interaction Department of Computer Science, UT Dallas.

Technical Skills

- **Fabrication & Prototyping:** soft-robotics fabrication (silicone molding/casting), SLA (resin) & FDM 3D printing, OpenSCAD / CAD.
- **Electronics & Embedded:** ESP32 / Arduino, soft pneumatic and DC-motor actuation, closed-loop sensor control, wireless (ESP-NOW).
- **Machine Learning & Vision:** custom object detection (YOLO / Ultralytics), on-device deployment (Core ML), OpenCV.
- **Programming:** Python, C/C++ (embedded), Swift/SwiftUI, JavaScript/p5.js, Java/Processing, R.
- **Design & Analysis:** Figma, Blender, statistical analysis in R.
- **Research Methods:** human-subjects studies (interviews, surveys, in-field evaluation), mixed methods, sensor-data analysis.

Presentations

Apr 2026	Poster & Talk — “From Intention to Action: An Invasive BCI-Driven Wearable Framework for Everyday Hand Assistance in SCI Patients” Everyday Wearable for Personalized Health and Well-Being Workshop, ACM CHI 2026, Barcelona, Spain.
2026	Talk & Poster — “PuffFab: Making Shape-Transformable Rice Paper for Playful Food Fabrication” ACM TEI 2026.
2025	Paper Presentation — “‘I Want to Use Technology to Help Farmers in the Future’” ACM Interaction Design and Children (IDC) 2025.

Awards

2023	Center for the Study of Contemporary China (CSCC) Fellowship Duke Kunshan University.
2022 – 2024	Dean’s List Duke Kunshan University (Fall 2022, Fall 2023, Spring 2024, Fall 2024).